

Practice Problems (1D Array)

1. Find the Maximum and Minimum Elements in an Array

Write a program to find and print the largest and smallest numbers in a given array.

- **Input:** An array of integers.
- **Output:** The maximum and minimum values in the array.
- **Logic:** Use a single loop to iterate through the array, comparing each element with the current maximum and minimum values.

2. Reverse the Elements of an Array

Write a program to reverse the order of elements in an array.

- **Input:** An array of integers.
- **Output:** The reversed array.
- **Logic:** Swap the first and last elements, then the second and second-last elements, and so on until the middle of the array is reached.

3. Check if an Array Is Sorted in Ascending Order

Write a program to determine if the elements of an array are sorted in non-decreasing order.

- **Input:** An array of integers.
- **Output:** YES if the array is sorted, otherwise NO.
- **Logic:** Traverse the array and check if each element is smaller than or equal to the next. If any element violates this condition, the array is not sorted.

4. Perform a Linear Search for a Value in an Array

Write a program to search for a specific value in an array and return its index (or -1 if not found).

- **Input:** An array of integers and a target value.
- **Output:** The index of the target value or -1 if it is not present.
- **Logic:** Traverse the array and compare each element with the target. Return the index when a match is found.

5. Find the Sum of All Elements in an Array

Write a program to compute the total sum of elements in an array.

- **Input:** An array of integers.
- **Output:** The sum of all elements.
- **Logic:** Use a loop to traverse the array, adding each element to a running total.

6. Find the Largest Difference Between Any Two Elements in an Array

Write a program to calculate the maximum absolute difference between any two elements in an array.

- **Input:** An array of integers.
- **Output:** The largest difference (absolute value).
- **Logic:** Find the minimum and maximum values in the array. The largest difference will be $\text{max} - \text{min}$.

7. Print the Number of Distinct Elements in an Array Already Sorted in Ascending Order

Problem:

Given a sorted array of integers, count and print the number of distinct elements in the array.

- **Input:** A sorted array of integers, e.g., [1, 1, 2, 2, 3, 4, 4, 5].
- **Output:** The number of distinct elements, e.g., 5 (elements: 1, 2, 3, 4, 5).

8. Rotate an Array to the Left by k Positions

Problem:

Write a program to rotate the elements of an array to the left by k positions. After rotation, the first k elements move to the end of the array, and the rest shift to the left.

- **Input:**
 - An array of integers, e.g., [1, 2, 3, 4, 5].
 - An integer k , e.g., 2.
- **Output:**
 - The rotated array, e.g., [3, 4, 5, 1, 2].